



St. MICHAEL

COLLEGE OF ENGG. & TECH.

(An ISO 9001:2015 Certified Institution)

(Approved by AICTE, New Delhi. & Affiliated to Anna University, Chennai.)

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Ln. Dr. V. Michael
Founder



Academic Year : 2024-2025

Semester : 06

CCS374-WEB APPLICATION SECURITY LAB MANUAL



REGULATION – 2021

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

ANNA UNIVERSITY: CHENNAI – 600 025

Prepared by,

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BONAFIDE CERTIFICATE

This is to certified that this is a bonafide record of work done by the Following Student in
the _____ - _____
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1. Name of the Student : _____
2. Roll No / Reg No : _____
3. Branch : _____
4. Semester : _____

Signature of the Staff In-charge

Signature of the HOD

Submitted for the Practical Examination on held

Internal Examiner

External Examiner

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EX.NO:1(a) Install wireshark and explore the various protocols
(Analyze the difference between HTTP vs HTTPS)

Aim:

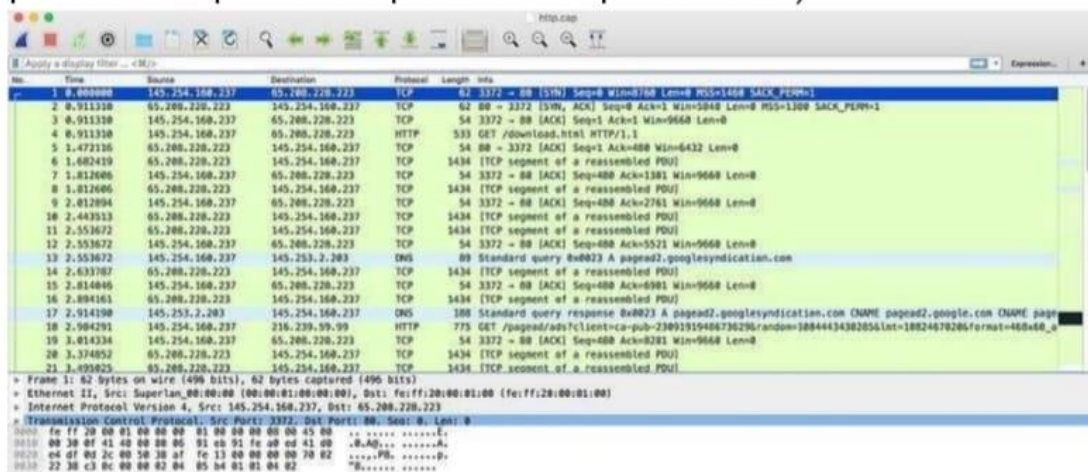
To explore the Analyze the difference between HTTP vs HTTPS by using wireshark

Installation of Wireshark Software

- Open the web browser.
- Search for 'Download Wireshark.'
- Select the Windows installer according to your system configuration, either 32-bit or 64-bit. Save the program and close the browser.
- Now, open the software, and follow the install instruction by accepting the license.
- The Wireshark is ready for use.

HTTP capture

Before start analyzing any packet, please turn off “Allow subdissector to reassemble TCP streams”(Preference → Protocol → TCP)(This will prevent TCP packet to split into multiple PDU unit)



As you can see I am using HTTP so that the encryption will not be hidden behind TLS.

As you can see at line number 13 standard DNS resolution is happening.

13	2.553072	145.254.160.237	145.254.160.237	DNS	89 Standard query 0x0023 A pagead2.googleadsyndication.com
14	2.633787	65.288.228.223	145.254.160.237	TCP	1434 [TCP segment of a reassembled PDU]
15	2.814846	145.254.160.237	65.288.228.223	TCP	54 3372 → 80 [ACK] Seq=488 Ack=6981 Win=9668 Len=0
16	2.894161	65.288.228.223	145.254.160.237	TCP	1434 [TCP segment of a reassembled PDU]
17	2.914198	145.253.2.203	145.254.160.237	DNS	188 Standard query response 0x0023 A pagead2.googleadsyndication.com CNAME pagead2.google.com CNAME pagead2.google.com
18	2.984291	145.254.160.237	216.239.59.99	HTTP	775 GET /pagead/ads/client-ca-pub-23891319486736296/random=180444343828561nt=18824678286format=488x68_u
19	3.014334	145.254.160.237	65.288.228.223	TCP	54 3372 → 80 [ACK] Seq=488 Ack=8281 Win=9668 Len=0
20	3.374852	65.288.228.223	145.254.160.237	TCP	1434 [TCP segment of a reassembled PDU]
21	3.492825	65.288.228.223	145.254.160.237	TCP	1434 [TCP segment of a reassembled PDU]

User Datagram Protocol, Src Port: 8089, Dst Port: 53
 Domain Name System (query)
 Transaction ID: 0x0023
 Flags: 0x0100 Standard query
 Questions: 1
 Answer RRs: 0
 Authority RRs: 0
 Additional RRs: 0
 Queries:

In line number 17 you see the response we are getting back with full DNS resolution

17	2.914198	145.253.2.203	145.254.160.237	DNS	188 Standard query response 0x0023 A pagead2.googleadsyndication.com CNAME pagead2.google.com CNAME pagead2.google.com
18	2.984291	145.254.160.237	216.239.59.99	HTTP	775 GET /pagead/ads/client-ca-pub-23891319486736296/random=180444343828561nt=18824678286format=488x68_u
19	3.014334	145.254.160.237	65.288.228.223	TCP	54 3372 → 80 [ACK] Seq=488 Ack=8281 Win=9668 Len=0
20	3.374852	65.288.228.223	145.254.160.237	TCP	1434 [TCP segment of a reassembled PDU]
21	3.492825	65.288.228.223	145.254.160.237	TCP	1434 [TCP segment of a reassembled PDU]

Transaction ID: 0x0023
 Flags: 0x0100 Standard query response, No error
 Questions: 1
 Answer RRs: 4
 Authority RRs: 0
 Additional RRs: 0
 Queries:
 -> pagead2.googleadsyndication.com: type A, class IN
 Answers:
 -> pagead2.googleadsyndication.com: type CNAME, class IN, cname pagead2.google.com
 -> pagead2.google.com: type CNAME, class IN, cname pagead.google-akads.net
 -> pagead.google-akads.net: type A, class IN, addr 216.239.59.104
 -> pagead.google-akads.net: type A, class IN, addr 216.239.59.99

Now if you look at Packet number 4 i.e is get request, HTTP primarily used two command

1: GET: To retrieve information

2: POST: To send information(For eg: when we submit some form we fill some data i.e is POST)

4	0.911350	145.254.160.237	65.288.228.223	HTTP	533 GET /download.html HTTP/1.1
5	1.472116	65.288.228.223	145.254.160.237	TCP	54 80 → 3372 [ACK] Seq=1 Ack=488 Win=6432 Len=0
6	1.682419	65.288.228.223	145.254.160.237	HTTP	1434 HTTP/1.1 200 OK
7	1.812606	145.254.160.237	65.288.228.223	TCP	54 3372 → 80 [ACK] Seq=488 Ack=1381 Win=9668 Len=0
8	1.812606	65.288.228.223	145.254.160.237	HTTP	1434 Continuation
9	2.012894	145.254.160.237	65.288.228.223	TCP	54 3372 → 80 [ACK] Seq=488 Ack=2761 Win=9668 Len=0
10	2.442553	65.288.228.223	145.254.160.237	HTTP	1434 Continuation

Frame 4: 533 bytes on wire (4264 bits), 533 bytes captured (4264 bits)
 Ethernet II, Src: Superlan_08:00:00:00:00:00, Dst: fe:ff:28:00:01:00 (fe:ff:28:00:01:00)
 Internet Protocol Version 4, Src: 145.254.160.237, Dst: 65.288.228.223
 Transmission Control Protocol, Src Port: 3372, Dst Port: 80, Seq: 1, Ack: 1, Len: 479
 Hypertext Transfer Protocol
 GET /download.html HTTP/1.1/r/n
 Host: www.ethereal.com/r/n
 User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.6) Gecko/20041113/r/n
 Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,image/png,image/jpeg,image/gif;q=0.2,*/*;q=0.1/r/n
 Accept-Language: en-us,en;q=0.5/r/n
 Accept-Encoding: gzip,deflate/r/n
 Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7/r/n
 Keep-Alive: 300/r/n
 Connection: keep-alive/r/n
 Referer: http://www.ethereal.com/development.html/r/n
 /r/n
 [Full request URI: http://www.ethereal.com/download.html]
 [HTTP request 1/1]
 [Response in frame 6]

Here I am trying to get download.html via HTTP protocol 1.1(The new version of protocol is now available i.e 2.0)

Then at line number 5 we see the acknowledgment as well as line number 6 server was able to found that page and send HTTP status code 200.

If you want more info about HTTP status code

HTTP Status Codes

You will see some more info like for packet 6, like Server type is Apache, content type is HTML, how long is the content length is,

6	1.662419	65.208.228.223	145.254.168.237	HTTP	1434	HTTP/1.1 200 OK
7	1.812606	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=1381 Win=9660 Len=0
8	1.812606	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
9	2.012894	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=2761 Win=9660 Len=0
10	2.443513	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
Frame 6: 1434 bytes on wire (11472 bits), 1434 bytes captured (11472 bits) on interface 0 Ethernet II, Src: fe8f:20:00:01:00 (fe8f:20:00:01:00), Dst: Superlan_00:00:01:00:00:00 Internet Protocol Version 4, Src: 65.208.228.223, Dst: 145.254.168.237 Transmission Control Protocol, Src Port: 80, Dst Port: 3372, Seq: 1, Ack: 480, Len: 1300 Hypertext Transfer Protocol HTTP/1.1 200 OK\r\n Date: Thu, 13 May 2004 18:17:12 GMT\r\n Server: Apache/2.0.46\r\n Last-Modified: Tue, 20 Apr 2004 13:17:00 GMT\r\n ETag: "9a81a-4096-7c354b00"\r\n Accept-Ranges: bytes\r\n Content-Length: 18070\r\n Keep-Alive: timeout=15, max=100\r\n Connection: Keep-Alive\r\n Content-Type: text/html; charset=ISO-8859-1\r\n \r\n [HTTP response 1/1] [Time since request: 0.771109000 seconds] [Request in frame: 4] File Data: 1086 bytes						

Then you will see bunch of continuation that is due to TCP window where you don't get acknowledgement for each and every packet

7	1.812606	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=1381 Win=9660 Len=0
8	1.812606	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
9	2.012894	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=2761 Win=9660 Len=0
10	2.443513	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
11	2.553672	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
12	2.553672	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=5521 Win=9660 Len=0
13	2.553672	145.254.168.237	65.208.228.223	DNS	89	Standard query 8x8023 A pagead2.googlesyndication.com
14	2.633787	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
15	2.834846	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=6981 Win=9660 Len=0
16	2.894161	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
17	2.914198	145.254.168.237	65.208.228.223	DNS	188	Standard query response 8x8023 A pagead2.googlesyndication.com CNAME pagead2.google.com CNAME pagead2.google.com
18	2.984291	145.254.168.237	65.208.228.223	HTTP	775	GET /pagead/ads?client=ca-pub-2389191948673629&random=1884443436285610024670266format=68x60_a
19	3.014334	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=8281 Win=9660 Len=0
20	3.374852	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
21	3.495825	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
22	3.495825	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=11041 Win=9660 Len=0
23	3.635227	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
24	3.645241	216.239.59.99	145.254.168.237	TCP	54	80 → 3371 [ACK] Seq=1 Ack=722 Win=31460 Len=0
25	3.815486	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=12421 Win=9660 Len=0
26	3.915638	216.239.59.99	145.254.168.237	HTTP	1404	HTTP/1.1 200 OK (text/html)
27	3.955688	216.239.59.99	145.254.168.237	HTTP	234	Continuation
28	3.955688	145.254.168.237	216.239.59.99	TCP	54	3371 → 80 [ACK] Seq=722 Ack=1591 Win=8760 Len=0
29	4.185904	65.208.228.223	145.254.168.237	HTTP	1434	Continuation
30	4.216862	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=480 Ack=13801 Win=9660 Len=0

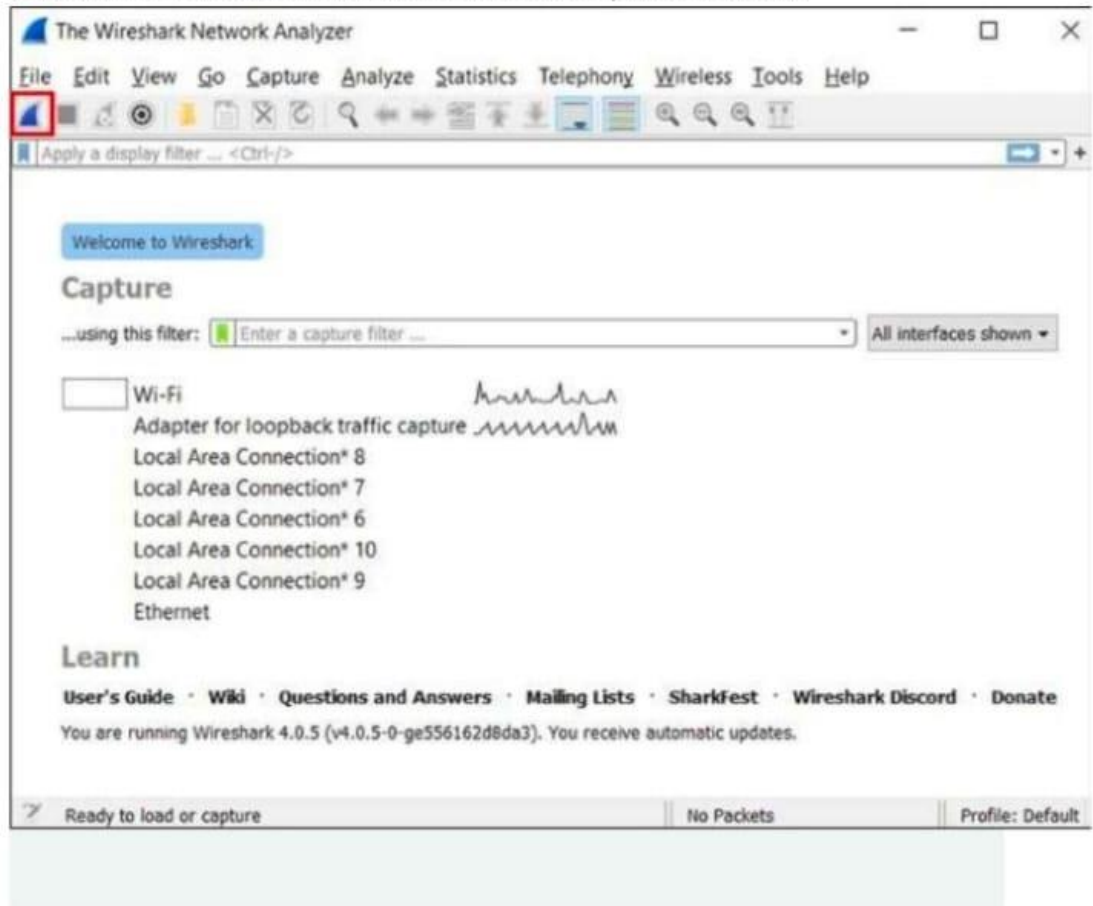
and at that top some usual TCP handshake

1	0.000000	145.254.168.237	65.208.228.223	TCP	62	3372 → 80 [SYN] Seq=0 Win=0 Len=0 MSS=1460 SACK_PERM=1
2	0.011310	65.208.228.223	145.254.168.237	TCP	62	80 → 3372 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1300 SACK_PERM=1
3	0.011310	145.254.168.237	65.208.228.223	TCP	54	3372 → 80 [ACK] Seq=1 Ack=1 Win=9660 Len=0

capture HTTPS in Wireshark

Capture and Decrypt Session Keys

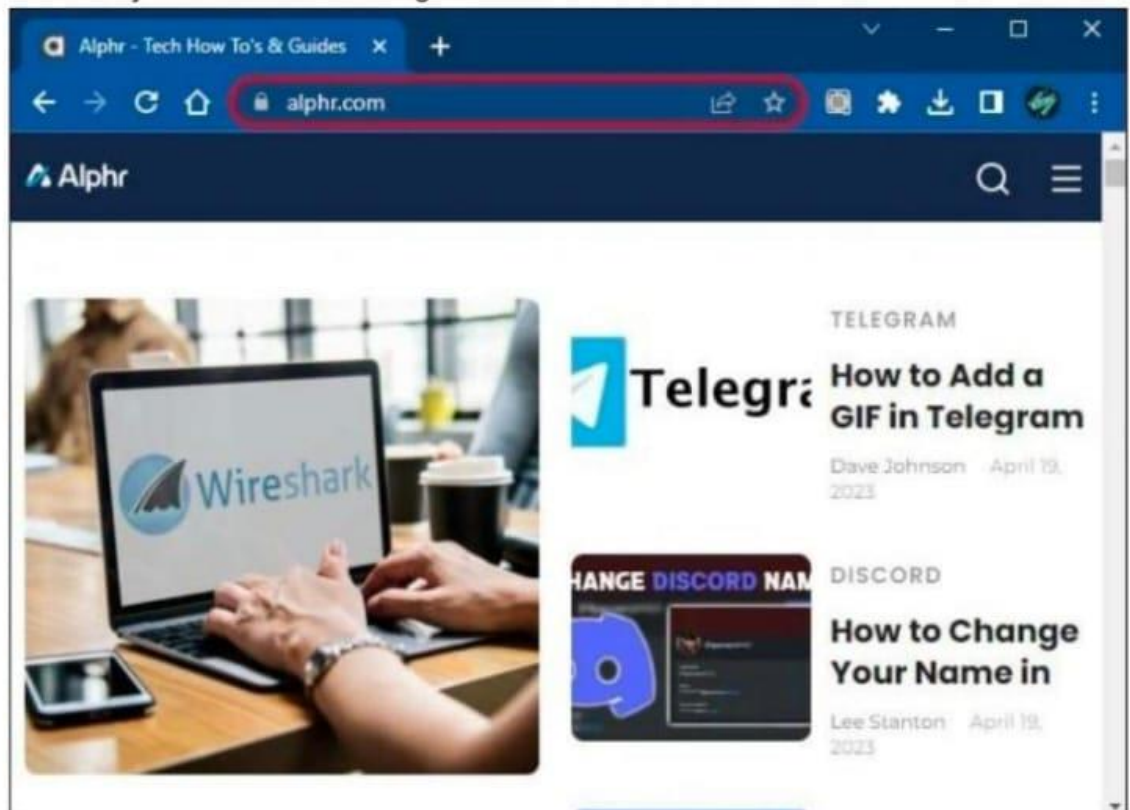
1. Launch Wireshark and start an unfiltered capture session.



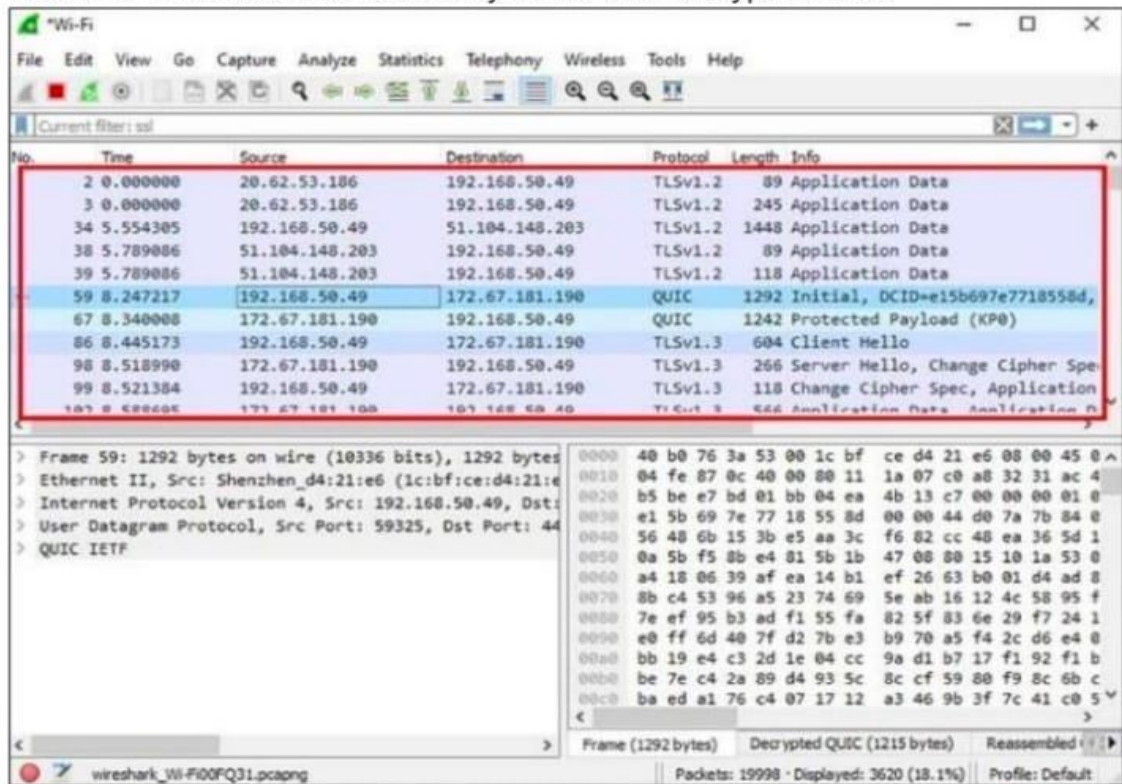
2. Minimize the Wireshark window and open your browser.



3. Go to any secure website to get data.



- Return to Wireshark and select any frame with encrypted data.



- Find "Packet byte view" and look at "Decrypted SSL" data. HTML should now be visible.

Conclusion

HTTPS provides better security than HTTP due to the following reasons:

- **Encryption:** The use of SSL/TLS encryption in HTTPS ensures that the data being transmitted between the client and the server is secure and cannot be intercepted by an attacker.
- **Authentication:** HTTPS uses digital certificates to authenticate the identity of the website, making it harder for attackers to impersonate a website and carry out a man-in-the-middle attack.
- **Integrity:** HTTPS provides data integrity which ensures that the data being transmitted between the client and the server has not been modified or tampered with.
- **SEO:** HTTPS is now a ranking signal for search engines, and having an HTTPS site can improve your search engine ranking.

Result

EX.NO:1(b) Install wireshark and explore the various protocols
 (Analyze the various security mechanisms embedded
 with different protocols)

Aim:

To Analyze the various security mechanisms embedded with different protocols by using wireshark

Analyzing the security mechanisms embedded in different protocols using Wireshark involves examining network traffic captures to understand how various security features are implemented and whether they are effective. Below are some common security mechanisms embedded in different protocols that you can analyze using Wireshark:

1. Transport Layer Security (TLS/SSL):

- TLS/SSL is used to encrypt and secure data transmission over the network.
- In Wireshark, you can identify TLS/SSL traffic by looking at the protocol field, which should indicate TLS or SSL.
- You can analyze the TLS handshake process, certificate exchange, and the encryption ciphers being used.

2. Secure Shell (SSH):

- SSH is a secure protocol for remote access and data exchange.
- Wireshark can capture SSH traffic, and you can analyze the key exchange, authentication methods, and encryption algorithms being used.

3. IPsec (Internet Protocol Security):

- IPsec is used to secure IP communication through encryption and authentication.
- Wireshark can capture and analyze IPsec packets to understand the negotiation of security associations (SAs), encryption algorithms, and authentication methods.

4. HTTP Secure (HTTPS):

- HTTPS is a secure version of HTTP that uses TLS/SSL for encryption.
- Wireshark can capture HTTPS traffic and allow you to examine the encrypted content after decryption by examining the TLS layer.

5. Virtual Private Network (VPN) Protocols:

- VPN protocols like OpenVPN, L2TP, or PPTP provide secure tunnels for data transmission.
- Wireshark can capture VPN traffic, and you can analyze the encapsulation, authentication, and encryption mechanisms used within the VPN tunnel.

6. Secure Sockets Layer (SSL) VPN:

- SSL VPNs create secure connections for remote access.
- Wireshark can capture SSL VPN traffic and allow you to analyze the SSL handshake, certificate exchange, and data transmission.

7. Kerberos:

- Kerberos is a network authentication protocol.
- You can capture and analyze Kerberos traffic in Wireshark to observe the ticket granting process, authentication, and encryption.

8. SNMPv3 (Simple Network Management Protocol version 3):

- SNMPv3 includes security features such as authentication and encryption.
- Wireshark can help you understand how SNMPv3 is used to secure network management.

9. DNSSEC (Domain Name System Security Extensions):

- DNSSEC adds security to the DNS by signing DNS records.
- Wireshark can be used to capture DNSSEC-signed DNS queries and responses to validate the digital signatures.

10. OAuth and OAuth2:

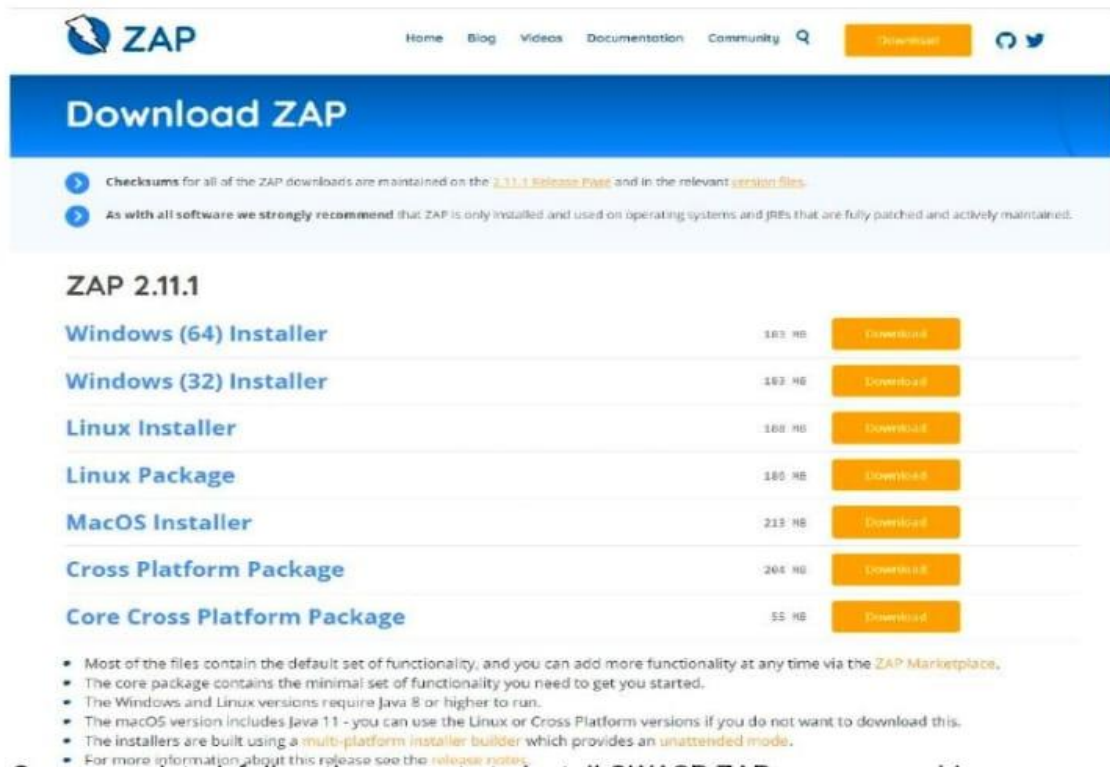
- OAuth is used for delegated authorization, often in web and API interactions.
- You can capture OAuth flows and analyze the token exchange and authentication mechanisms.

When analyzing security mechanisms using Wireshark, it's essential to focus on packet captures related to the specific protocol or technology you're interested in. Look for indicators of encryption, authentication, key exchanges, and other security features to assess the effectiveness and security of the implementation. Additionally, pay attention to any anomalies or potential vulnerabilities that may be present in the captured traffic.

EX.NO:2 Identify the vulnerabilities using OWASP ZAP tool

1. Installing ZAP

You can download the latest version from the OWASP ZAP website for your operating system to install ZAP or reference the [ZAP docs](#) for a more detailed installation guide.



The screenshot shows the OWASP ZAP website's download page. At the top, there's a navigation bar with links for Home, Blog, Videos, Documentation, and Community, along with a search icon and a 'Download' button. Below the navigation bar is a large blue banner with the text 'Download ZAP'. Underneath the banner, there are two informational points: 'Checksums for all of the ZAP downloads are maintained on the [2.11.1 Release Page](#) and in the relevant [version files](#).' and 'As with all software we strongly recommend that ZAP is only installed and used on operating systems and JREs that are fully patched and actively maintained.' Below this, the section 'ZAP 2.11.1' lists various download options with their respective sizes and 'Download' buttons. The options are: Windows (64) Installer (163 MB), Windows (32) Installer (163 MB), Linux Installer (168 MB), Linux Package (180 MB), MacOS Installer (213 MB), Cross Platform Package (264 MB), and Core Cross Platform Package (55 MB). At the bottom, there are several bullet points providing additional information about the downloads, including the availability of the ZAP Marketplace, the minimal set of functionality in the core package, the Java version requirements for Windows and Linux, the inclusion of Java 11 in the macOS version, the use of a multi-platform installer builder, and a link to the release notes for more information.

Download Option	Size	Download Button
Windows (64) Installer	163 MB	Download
Windows (32) Installer	163 MB	Download
Linux Installer	168 MB	Download
Linux Package	180 MB	Download
MacOS Installer	213 MB	Download
Cross Platform Package	264 MB	Download
Core Cross Platform Package	55 MB	Download

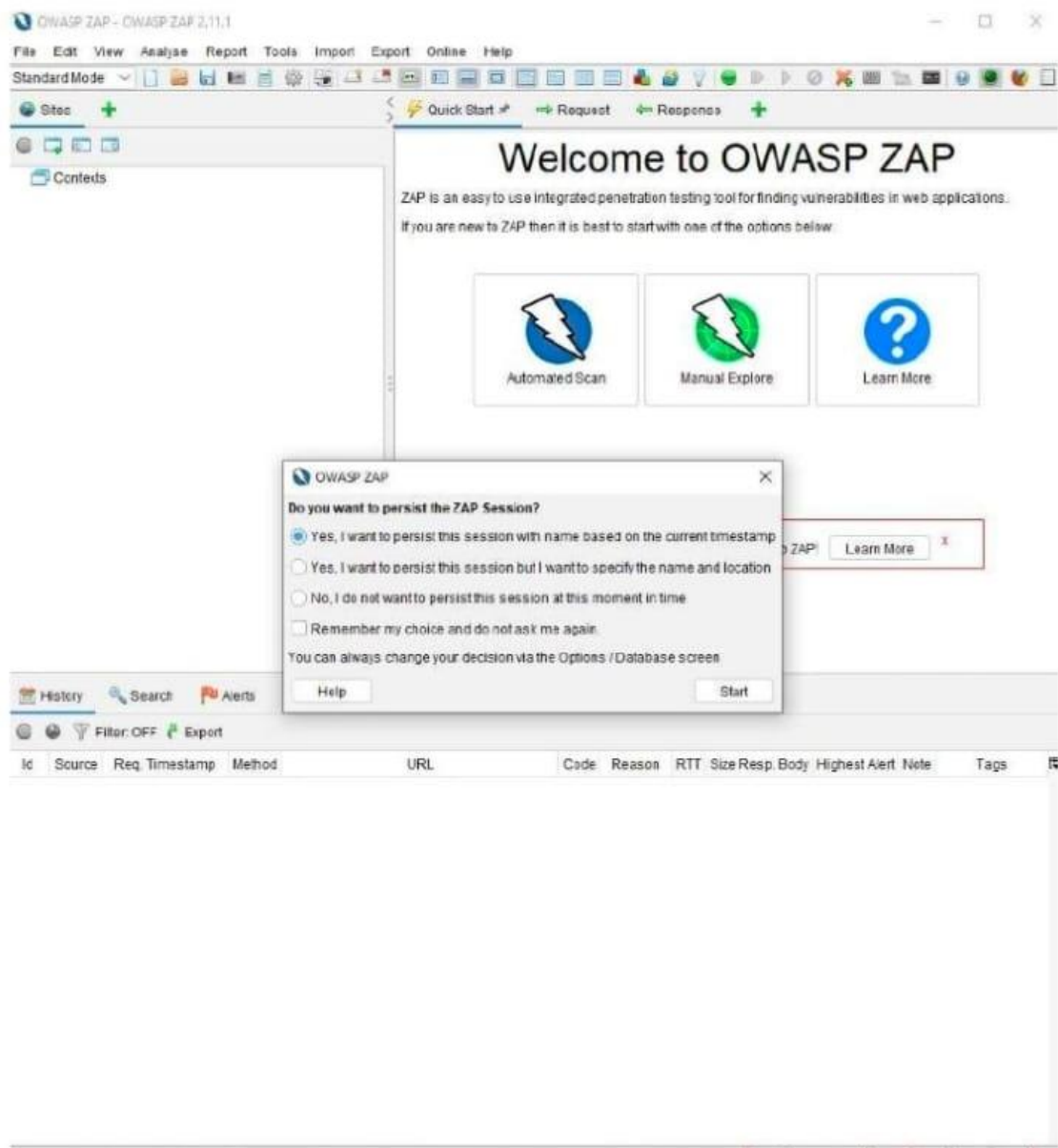
- Most of the files contain the default set of functionality, and you can add more functionality at any time via the [ZAP Marketplace](#).
- The core package contains the minimal set of functionality you need to get you started.
- The Windows and Linux versions require Java 8 or higher to run.
- The macOS version includes Java 11 - you can use the Linux or Cross Platform versions if you do not want to download this.
- The installers are built using a [multi-platform installer builder](#) which provides an [unattended mode](#).
- For more information about this release see the [release notes](#).

Once completed, follow the prompts to install OWASP ZAP on your machine.

2. Persisting a session

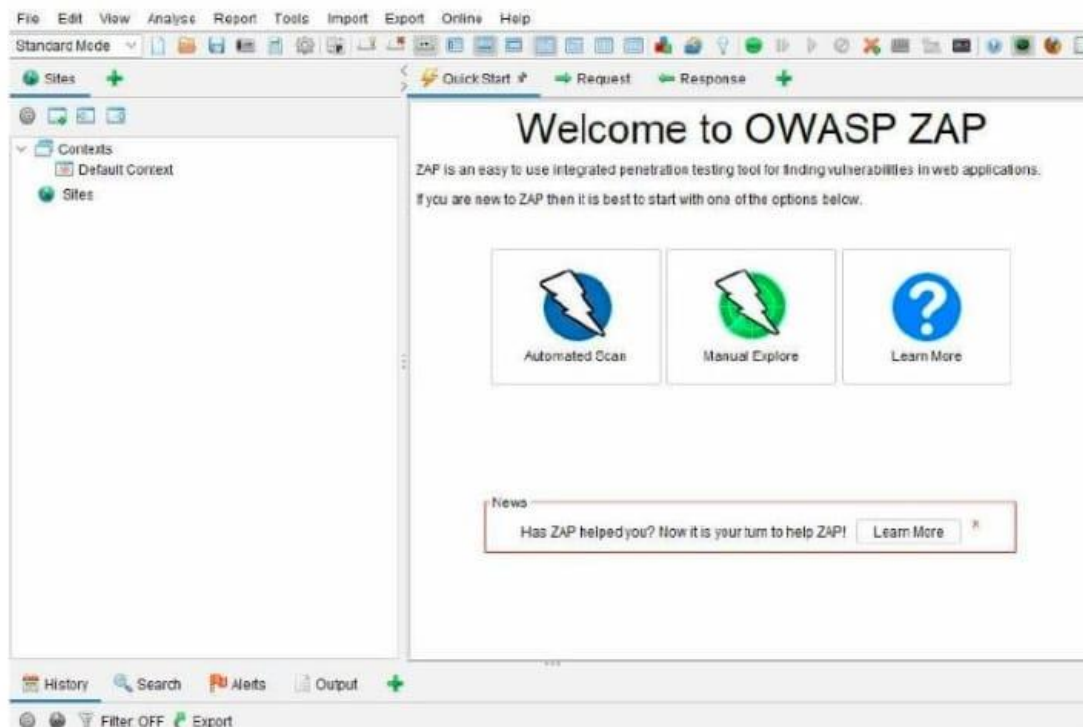
Persisting a session in OWASP ZAP means that the session will be saved and can be reopened at a later time. This is useful if you want to continue testing a website or application at a later time.

Once you've started OWASP ZAP, you will see a screen that looks like this:

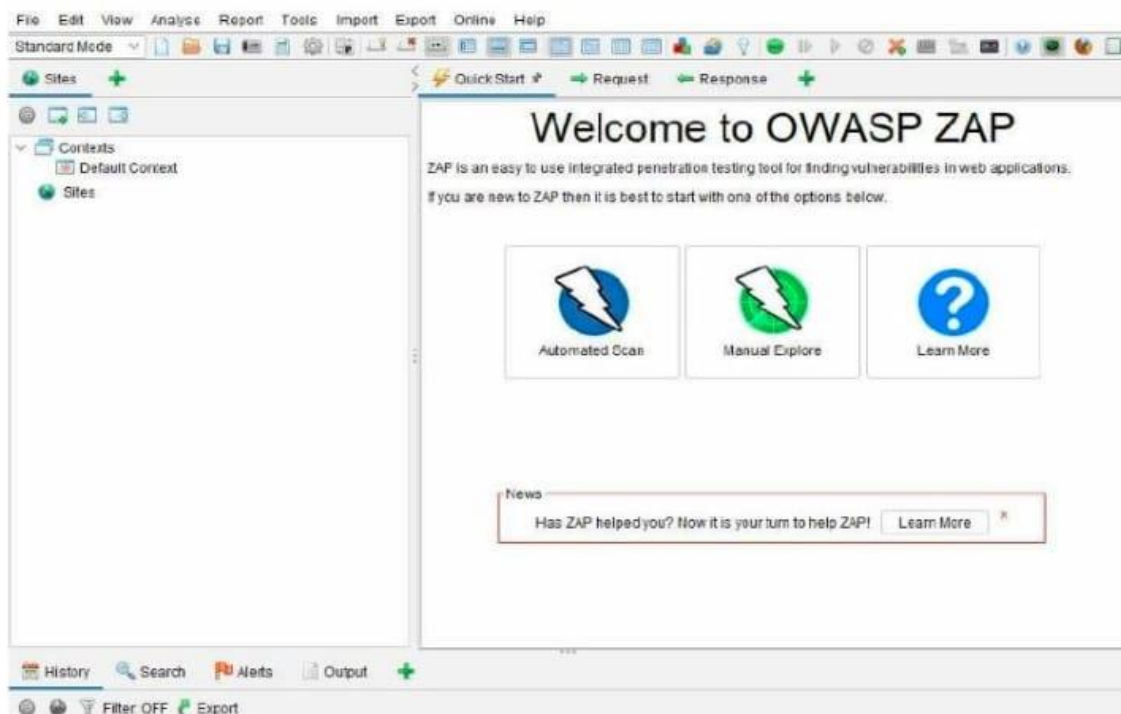


The prompt gives two options to persist in the session. You can use the default to name the session based on the current timestamp or set your name and location.

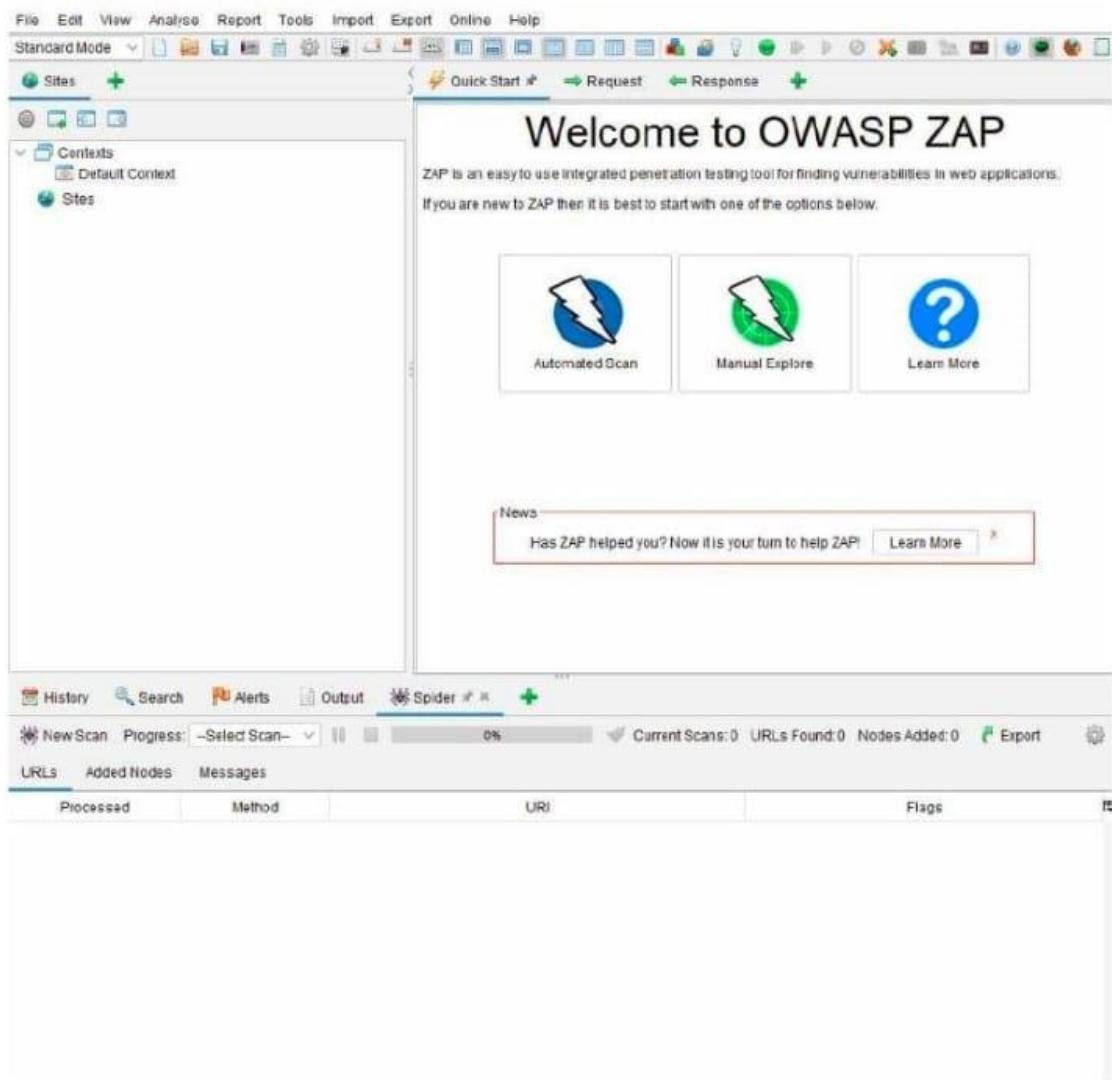
Alternatively, you can persist a session by going to 'File' and choosing 'Persist Session...'. Give your session a name and click on the 'Save' button.



To run an automated scan, you can use the quick start “Automated Scan” option under the “Quick Start” tab. Enter the URL of the site you want to scan in the “URL to attack” field, and then click “Attack!”.



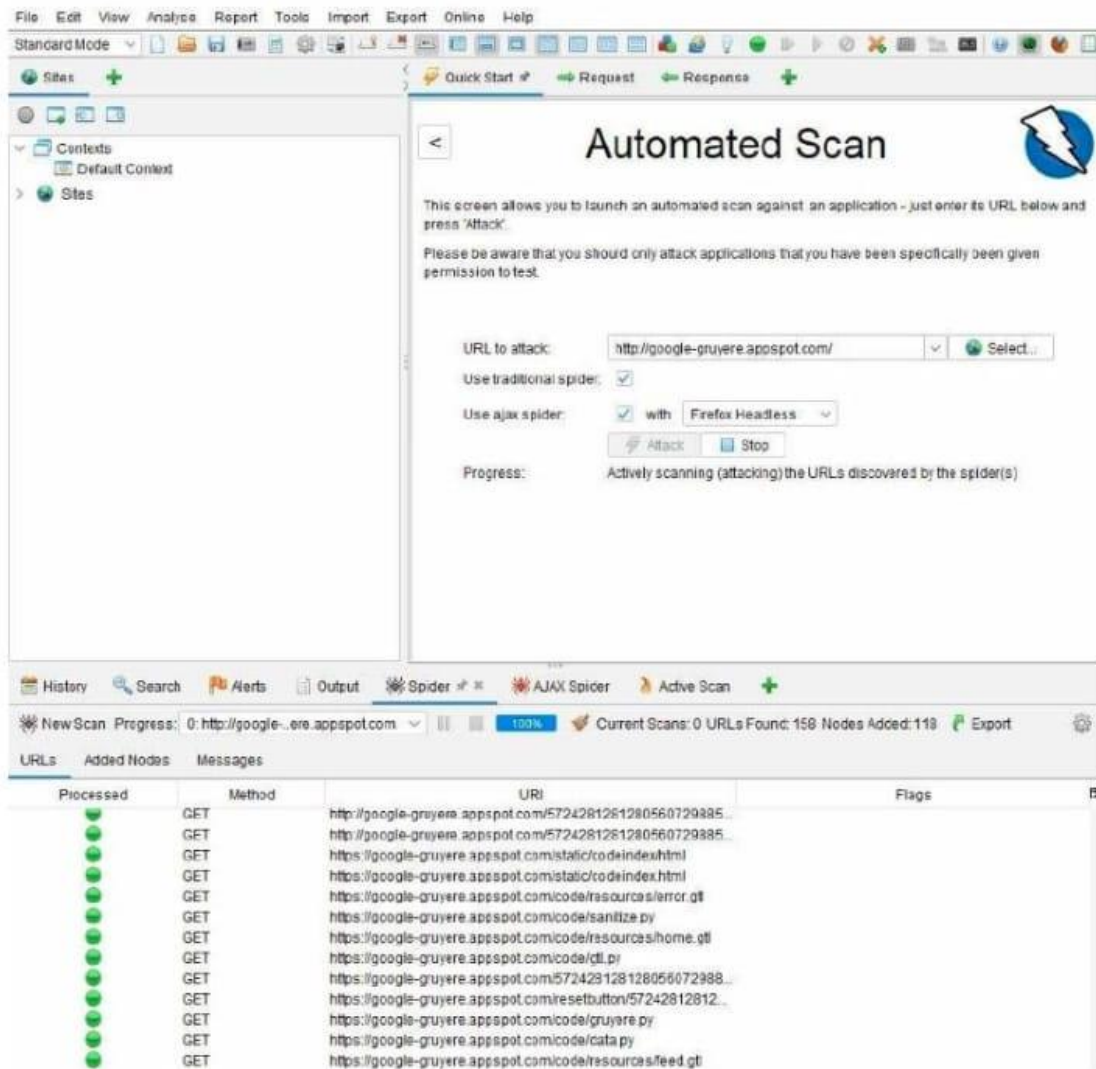
To run an automated scan, you can use the quick start “Automated Scan” option under the “Quick Start” tab. Enter the URL of the site you want to scan in the “URL to attack” field, and then click “Attack!”.



4. Interpreting test results

Interpreting test results in OWASP ZAP is vital to understand the scan findings and determine which issues require further investigation. Additionally, it can help to prioritize remediation efforts.

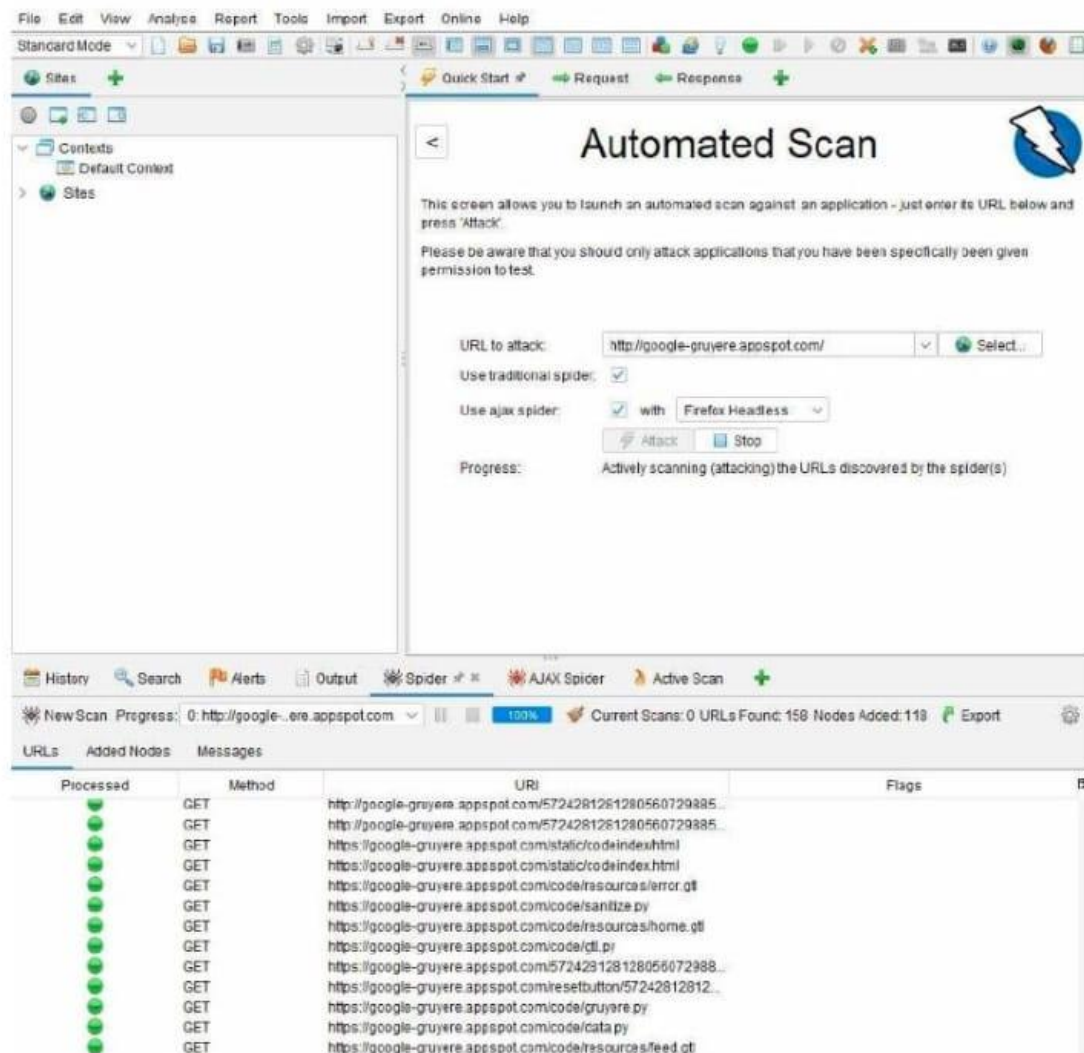
In OWASP ZAP, you can view alerts by clicking on the "Alerts" tab. This tab will show you a list of all the alerts that have been triggered during your testing. The alerts are sorted by risk level, with the highest risk alerts at the top of the list. OWASP ZAP will give details of the discovered vulnerabilities and suggestions on how you can fix them.



5. Viewing alerts and alert details

Viewing alerts and alert details in OWASP ZAP is a way to see what potential security issues have been identified on a website. It can help security and administrators understand what needs to be fixed to improve the app's security.

If you cannot find your 'Alerts' tab, you can access it via the 'View' menu, along with other options available in OWASP ZAP. Once you have your 'Alerts' tab, you can navigate the various vulnerabilities discovered and explore the reports generated by OWASP ZAP.



5. Viewing alerts and alert details

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EX.NO:3 Create simple REST API using python for following operation (GET,PUSH,POST,DELETE)

Aim:

To Create simple REST API using python for CRUD operation.

Procedure:

- Install Flask:
- Creating a simple REST API in Python for basic CRUD (Create, Read, Update, Delete) operations typically involves using a web framework.

Code:

1.Install Flask:

```
pip install flask
```

2.Create a new file called app.py:

```
from flask import Flask, request, jsonify
```

```
app = Flask(__name__)
```

```
# Sample data (for demonstration purposes)
```

```
data = [
```

```
    {"id": 1, "name": "Item 1", "description": "This is the first item."},
```

```
    {"id": 2, "name": "Item 2", "description": "This is the second item."},
```

```
    {"id": 3, "name": "Item 3", "description": "This is the third item."}
```

```
]
```

```
# GET (Retrieve all items)
```

```
@app.route('/items', methods=['GET'])
```

```
def get_items():
```

```
    return jsonify(data)
```

```
# POST (Create a new item)
```

```
@app.route('/items', methods=['POST'])
```

```
def create_item():
```

```
    item = request.get_json()
```

```
    data.append(item)
```

```

        return jsonify({"message": "Item created successfully"})

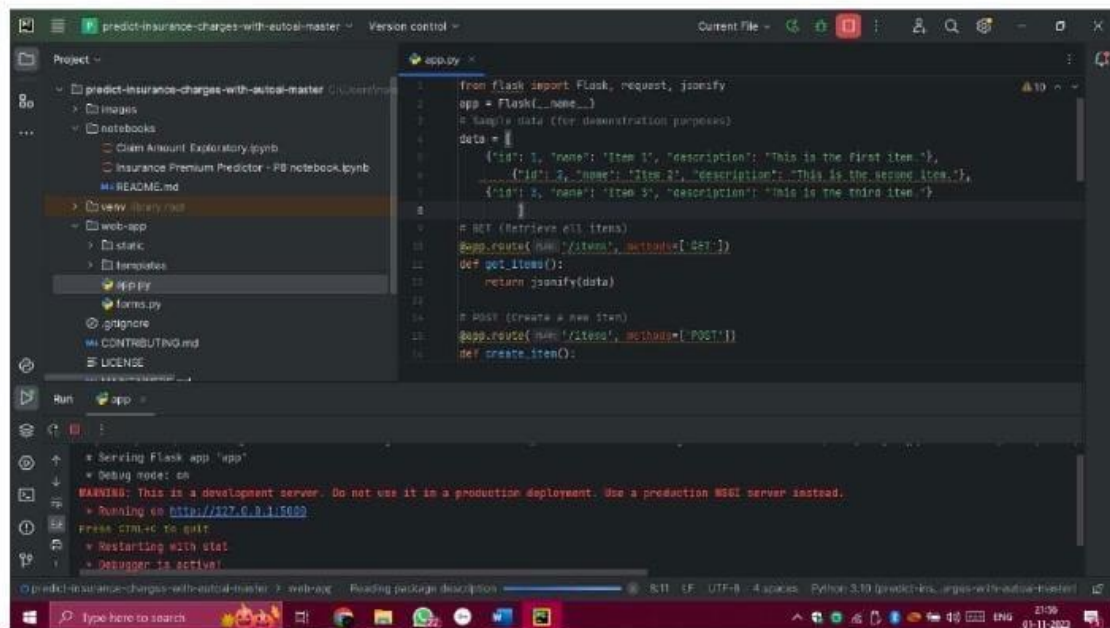
# PUT (Update an item)
@app.route('/items/<int:item_id>', methods=['PUT'])
def update_item(item_id):
    if 0 <= item_id < len(data):
        updated_item = request.get_json()
        data[item_id] = updated_item
        return jsonify({"message": "Item updated successfully"})
    else:
        return jsonify({"message": "Item not found"}, 404)

# DELETE (Delete an item)
@app.route('/items/<int:item_id>', methods=['DELETE'])
def delete_item(item_id):
    if 0 <= item_id < len(data):
        del data[item_id]
        return jsonify({"message": "Item deleted successfully"})
    else:
        return jsonify({"message": "Item not found"}, 404)

if __name__ == '__main__':
    app.run(debug=True)

```

Output:



The screenshot shows a VS Code editor with a project named 'predict-insurance-charges-with-autosai-master'. The file explorer on the left shows a directory structure with 'app.py' selected. The main editor displays the code for 'app.py', which is a Flask application. The terminal at the bottom shows the output of running the application, indicating it is serving on http://127.0.0.1:5000.

```
1 from flask import Flask, request, jsonify
2 app = Flask(__name__)
3 # Sample data (for demonstration purposes)
4 data = [
5     {'id': 1, 'name': 'Item 1', 'description': 'This is the first item.'},
6     {'id': 2, 'name': 'Item 2', 'description': 'This is the second item.'},
7     {'id': 3, 'name': 'Item 3', 'description': 'This is the third item.'}
8 ]
9
10 # GET (retrieves all items)
11 @app.route('/api/items', methods=[GET])
12 def get_items():
13     return jsonify(data)
14
15 # POST (create a new item)
16 @app.route('/api/items', methods=[POST])
17 def create_item():
```

Run app

Serving Flask app 'app'

Debug mode: on

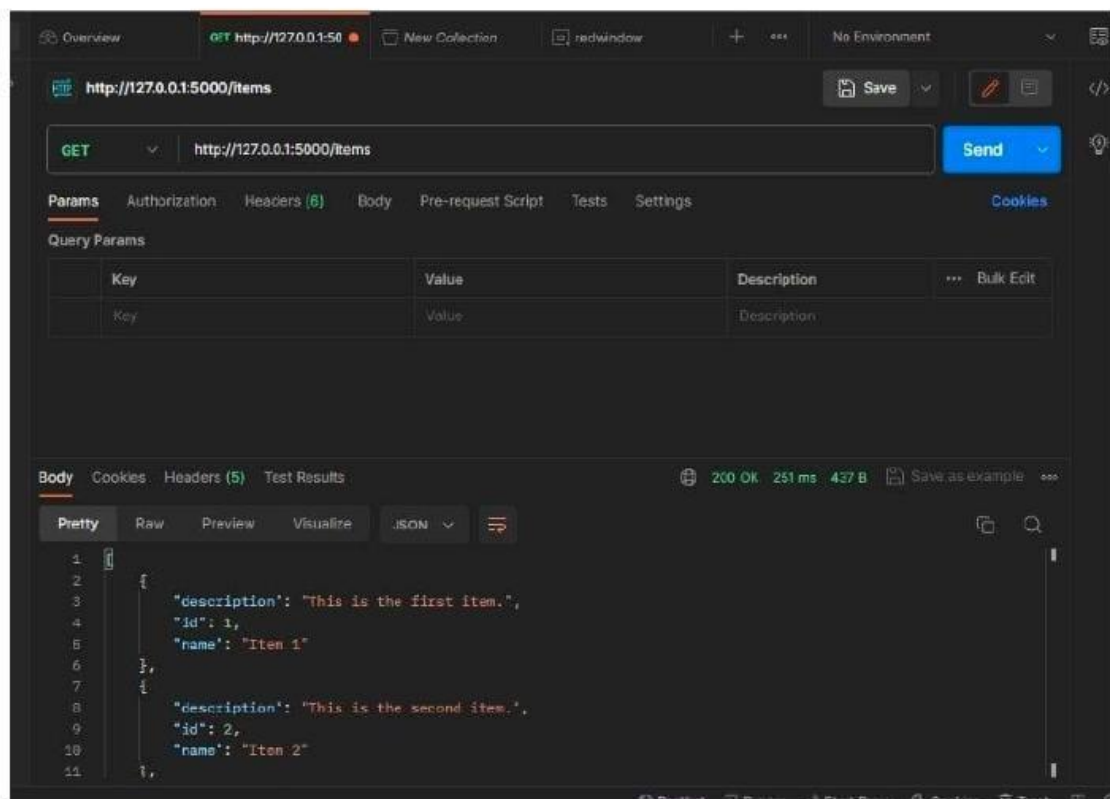
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.

Running on http://127.0.0.1:5000

press CTRL+C to quit

Restarting with stat

Debugger is active!



The screenshot shows a REST client interface with a GET request to 'http://127.0.0.1:5000/items'. The response is a JSON array of three items. The 'Body' tab is selected, showing the JSON response in a pretty-printed format.

Overview GET http://127.0.0.1:5000/items

Save Send

Params Authorization Headers (6) Body Pre-request Script Tests Settings Cookies

Query Params

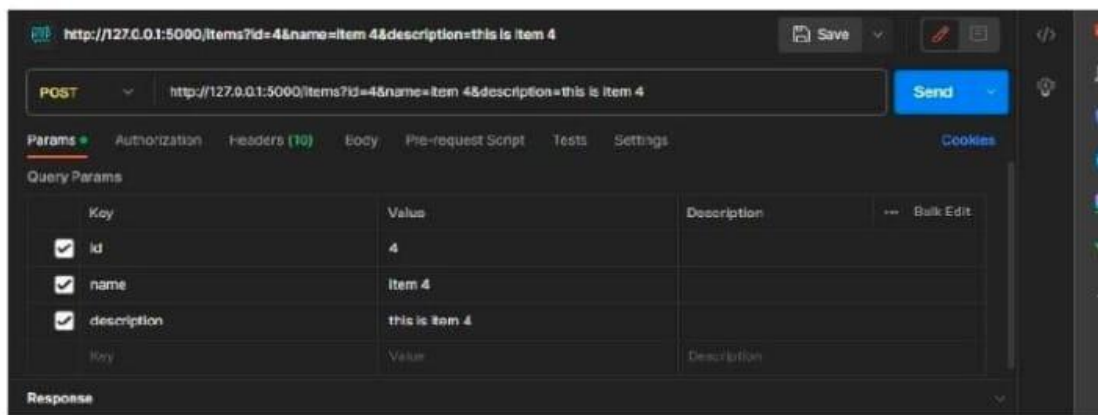
Key	Value	Description
Key	Value	Description

Body Cookies Headers (5) Test Results

200 OK 251 ms 437 B Save as example

Pretty Raw Preview Visualize JSON

```
1 {
2   "description": "This is the first item.",
3   "id": 1,
4   "name": "Item 1"
5 },
6 {
7   "description": "This is the second item.",
8   "id": 2,
9   "name": "Item 2"
10 },
11 {
```



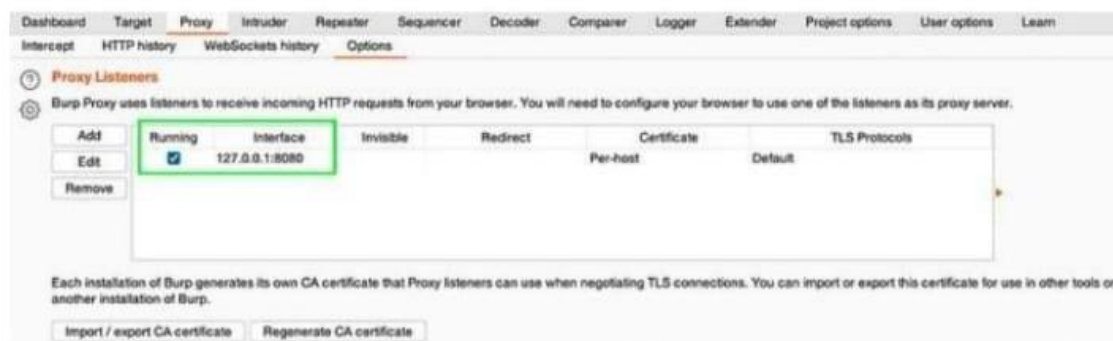
EX.NO:4 Install Burp Suite to do following vulnerabilities

Aim

To Testing for SQL injection vulnerabilities with Burp Suite

Prepare the Burp Suite

1. [Download](#) and install Burp Suite Community Edition;
2. Run Burp Suite Community Edition and choose on the start screen: Temporary project → [Next] → Use Burp defaults → [Start Burp]
3. Check Burp's proxy settings: Proxy → Options → Proxy Listeners. Burp's proxy should listen 127.0.0.1:8080



Procedure

SQL injection

1. Identify a request that you want to investigate.
2. In the request, highlight the parameter that you want to test and select Send to Intruder.
3. Go to the Intruder > Positions tab. Notice that the parameter has been automatically marked as a payload position.
4. Go to the Payloads tab. Under Payload settings [Simple list] add a list of SQL fuzz strings.

1. If you're using Burp Suite Professional, open the Add from list drop-down menu and select the built-in Fuzzing - SQL wordlist.
2. If you're using [Burp Suite Community Edition](#), manually add a list.
5. Under Payload processing, click Add. Configure payload processing rules to replace any list placeholders with an appropriate value. You need to do this if you're using the built-in wordlist:
 1. To replace the {base} placeholder, select Replace placeholder with base value.
 2. To replace other placeholders, select Match/Replace, then specify the placeholder and replacement. For example, replace {domain} with the domain name of the site you're testing.



6. Click Start attack. The attack starts running in a new dialog. Intruder sends a request for each SQL fuzz string on the list.
7. When the attack is finished, study the responses to look for any noteworthy behavior. For example, look for:
 - Responses that include additional data as a result of the query.
 - Responses that include other differences due to the query, such as a "welcome back" message or error message.
 - Responses that had a time delay due to the query.

If you're using the lab, look for responses with a longer length. These may include additional products.

Results	Positions	Payloads	Resource pool	Settings		
Filter: Showing all items						
Request	Payload	Status code	Error	Timeout	Length	Comment
44	Clothing%2c+shoes+and+accessories or 7=7	200			3395	
45	Clothing%2c+shoes+and+accessories or 7=7--	200			3398	
46	Clothing%2c+shoes+and+accessories or 7=7#	200			3396	
47	Clothing%2c+shoes+and+accessories or 7=7)--	200			3399	
48	Clothing%2c+shoes+and+accessories or 7=7)#	200			3397	
49	Clothing%2c+shoes+and+accessories' or 7=7	500			2323	
50	Clothing%2c+shoes+and+accessories' or 7=7--	200			11261	
51	Clothing%2c+shoes+and+accessories' or 7=7#	500			2323	
52	Clothing%2c+shoes+and+accessories' or 'z'='z	200			9336	
53	Clothing%2c+shoes+and+accessories' or 'z'='z' or 'a'='b	200			13124	
54	Clothing%2c+shoes+and+accessories'/' or '/'z'='z	200			11241	
55	Clothing%2c+shoes+and+accessories' or username like '%	500			4140	

cross-site scripting (XSS)

Bypassing XSS filters by enumerating permitted tags and attributes

1. In Proxy > HTTP history, right-click the request with a reflected input that you want to investigate. Select Send to Intruder.
2. Identify whether any tags are permitted:
 1. In Intruder, replace the value of the input with: <>.
 2. Click inside the angle brackets, then click Add § twice to add a payload position.

ⓘ Payload positions
Configure the positions where payloads will be inserted, they can be added into the target as well as the base request.

Target: ☒ Update Host header to match target

1 GET /?search=<> HTTP/2
2 Host: 0a59000c0331ed4980c4efde006500d3.web-security-academy.net
3 Cookie: session=vgyF395fMRiyk2jZkF7tEdZ6cJFbTdp2
4 Sec-Ch-Ua: "Chromium";v="113", "Not-A.Brand";v="24"
5 Sec-Ch-Ua-Mobile: 0
6 Sec-Ch-Ua-Platform: "macOS"

Add §
Clear §
Auto §
Refresh

3. Go to the Payloads tab. Under Payload settings [simple list] add a list of tags that you want to test. For example, use the tags [in the XSS cheat sheet](#).
4. Click Start attack. The attack starts running in a new dialog. Intruder sends a request for each tag on the list.

5. When the attack is finished, look for any responses with a 200 status code. This indicates that the tag is permitted. If a tag is filtered out, it has a 400 status code instead.
3. Identify whether any attributes are permitted:
 1. In Intruder > Positions, update the payload position. Add a tag that you enumerated in the previous step, then add payload markers to test different attributes.

② Payload positions

Configure the positions where payloads will be inserted, they can be added into the target as well as the base request.

Target:

☒ Update Host header to match target

Add §

Clear §

Auto §

Refresh

```
1 GET /?search=body%20%3C%3E HTTP/2
2 Host: 0a59000c0331ed4980c4efde006500d3.web-security-academy.net
3 Cookie: session=vgyF395fMRlyk2jZkF7tEdZ6cJfbTdp2
4 Sec-Ch-Ua: "Chromium";v="113", "Not-A.Brand";v="24"
5 Sec-Ch-Ua-Mobile: ?0
6 Sec-Ch-Ua-Platform: "macOS"
```

2. Go to Intruder > Payloads. Click Clear to remove the list of tags that you tested in the previous step.
3. Under Payload settings [Simple list] add a list of attributes that you want to test. For example, use the attributes in the [XSS cheat sheet](#).
4. Click Start attack. The attack starts running in a new dialog. Intruder sends a request for each attribute on the list.
5. When the attack is finished, look for any responses with a 200 status code. This indicates that an attribute is permitted.

EX.NO:5 Attack the website using Social Engineering method

Aim

To Attack the website using Social Engineering method

Procedure

Step 1: First, we have to open the Kali Linux Terminal and move to Desktop.

1. cd Desktop

```
File Actions Edit View Help


```
(preeti@kali)~[~]
$ cd Desktop
```


```

Step 2: Now, we are on a desktop so use the following command in order to create a new directory called SEToolkit.

1. mkdir SEToolkit

```
(preeti@kali)~[~/Desktop]
$ mkdir SEToolkit



```
(preeti@kali)~[~/Desktop]
$
```


```

Step 3: Now, we are in the Desktop directory though we have created a SEToolkit directory so go to SEToolkit directory using the following command.

1. cd SEToolkit

```
(preeti@kali)~[~/Desktop]
$ cd SEToolkit



```
(preeti@kali)~[~/Desktop/SEToolkit]
$
```


```

Step 4: Now we're in the SEToolkit directory, we will need to clone SEToolkit from GitHub in order to utilize it.

1. `git clone https://github.com/trustedsec/social-engineer-toolkit setoolkit/`

```
(preeti@kali)-[~/Desktop]
$ cd SEToolkit

(preeti@kali)-[~/Desktop/SEToolkit]
$ git clone https://github.com/trustedsec/social-engineer-toolkit setoolkit/
Cloning into 'setoolkit' ...
remote: Enumerating objects: 110242, done.
remote: Counting objects: 100% (180/180), done.
remote: Compressing objects: 100% (151/151), done.
remote: Total 110242 (delta 95), reused 63 (delta 28), pack-reused 110062
Receiving objects: 100% (110242/110242), 175.29 MiB | 9.77 MiB/s, done.
Resolving deltas: 100% (68354/68354), done.
```

Step 5: Now, the Social Engineering Toolkit has been downloaded to our directory, we have to use the following command in order to navigate to the social engineering toolkit's internal directory.

1. `cd setoolkit`

```
(preeti@kali)-[~/Desktop/SEToolkit]
$ cd setoolkit

(preeti@kali)-[~/Desktop/SEToolkit/setoolkit]
$ ls
Dockerfile  readme      requirements.txt  seproxy      setup.py  src
modules     README.md   seautomate       setoolkit    seupdate
```

Step 6: Now we have successfully downloaded the social engineering toolkit in our directory SEToolkit. Now we can use the following command to install the requirements.

1. `pip3 install -r requirements.txt`

```
(preeti@kali)-[~/Desktop/SEToolkit/setoolkit]
$ pip3 install -r requirements.txt
Requirement already satisfied: pexpect in /usr/lib/python3/dist-packages (from -r requirements.txt (line 1)) (4.8.0)
Collecting pycrypto
  Downloading pycrypto-2.6.1.tar.gz (446 kB)
    446 kB 4.1 MB/s
Requirement already satisfied: requests in /usr/lib/python3/dist-packages (from -r requirements.txt (line 3)) (2.25.1)
Requirement already satisfied: pyopenssl in /usr/lib/python3/dist-packages (from -r requirements.txt (line 4)) (20.0.1)
Requirement already satisfied: pefile in /usr/lib/python3/dist-packages (from -r requirements.txt (line 5)) (2019.4.18)
Requirement already satisfied: impacket in /usr/lib/python3/dist-packages (from -r requirements.txt (line 6)) (0.9.22)
Requirement already satisfied: qrcode in /usr/lib/python3/dist-packages (from -r requirements.txt (line 8)) (6.1)
Requirement already satisfied: pillow in /usr/lib/python3/dist-packages (from -r requirements.txt (line 9)) (8.1.0)
Requirement already satisfied: pymysql<3.0 in /usr/lib/python3/dist-packages (from -r requirements.txt (line 11)) (2.1.4)
Building wheels for collected packages: pycrypto
  Building wheel for pycrypto (setup.py) ... done
  Created wheel for pycrypto: filename=pycrypto-2.6.1-cp39-cp39-linux_x86_64.whl size=526405 sha256=7dcdde3a531c35a8d26af25586fcc0a1f45377dcd1cbeb4a15e576309f682de
  Stored in directory: /home/preeti/.cache/pip/wheels/9d/29/32/8b8f22481bec8b0f6e7067927336ec167faff2ed9db849448f
Successfully built pycrypto
Installing collected packages: pycrypto
Successfully installed pycrypto-2.6.1
```

Step 7: All the requirements have been downloaded to our setoolkit. Now it's time to install the requirements we have downloaded.

1. python setup.py

```
(preeti@kali)-[~/Desktop/SEToolkit/setoolkit]
$ python setup.py
[*] Installing requirements.txt ...
Requirement already satisfied: pexpect in /usr/lib/python3/dist-packages (from -r requirements.txt (line 1)) (4.8.0)
Requirement already satisfied: pycrypto in /home/preeti/.local/lib/python3.9/site-packages (from -r requirements.txt (line 2)) (2.6.1)
Requirement already satisfied: requests in /usr/lib/python3/dist-packages (from -r requirements.txt (line 3)) (2.25.1)
Requirement already satisfied: pyopenssl in /usr/lib/python3/dist-packages (from -r requirements.txt (line 4)) (20.0.1)
Requirement already satisfied: pefile in /usr/lib/python3/dist-packages (from -r requirements.txt (line 5)) (2019.4.18)
Requirement already satisfied: impacket in /usr/lib/python3/dist-packages (from -r requirements.txt (line 6)) (0.9.22)
Requirement already satisfied: qrcode in /usr/lib/python3/dist-packages (from -r requirements.txt (line 8)) (6.1)
Requirement already satisfied: pillow in /usr/lib/python3/dist-packages (from -r requirements.txt (line 9)) (8.1.0)
Requirement already satisfied: pymssql<3.0 in /usr/lib/python3/dist-packages (from -r requirements.txt (line 11)) (2.1.4)
[*] Installing setoolkit to /usr/share/setoolkit..
/home/preeti/Desktop/SEToolkit/setoolkit
mkdir: cannot create directory '/usr/share/setoolkit/': Permission denied
mkdir: cannot create directory '/etc/setoolkit/': File exists
cp: target '/usr/share/setoolkit/' is not a directory
cp: cannot create regular file '/etc/setoolkit/set.config': Permission denied
[*] Creating launcher for setoolkit ...
Traceback (most recent call last):
  File "setup.py", line 15, in <module>
    filewrite = open("/usr/local/bin/setoolkit", "w")
IOError: [Errno 13] Permission denied: '/usr/local/bin/setoolkit'
```

Step 8: Finally all the installation process is complete now it's time to run the Social Engineering Toolkit. We have to type the following command in order to run the SEToolkit.

1. Setoolkit

Step 9: At this point, setoolkit will ask us (y) or (n). When we type y, our social engineering toolkit will start running.

1. Y

```
File Actions Edit View Help

  \
  || Social-Engineer Toolkit ||
  ||      Free      ||
  ||     #hugs     ||
  ||   By: TrustedSec   ||
  /

/====/
/oooo oooo oooo oooo /
/ooooooooooooooooooooo/
/ooooooooooooooooooooo/
/C=_____

[---] The Social-Engineer Toolkit (SET) [---]
[---] Created by: David Kennedy (ReLlK) [---]
[---] Version: 0.0.3 [---]
[---] Codename: 'Maverick' [---]
[---] Follow us on Twitter: @TrustedSec [---]
[---] Follow me on Twitter: @HackingDave [---]
[---] Homepage: https://www.trustedsec.com [---]
Welcome to the Social-Engineer Toolkit (SET).
The one stop shop for all of your SE needs.

The Social-Engineer Toolkit is a product of TrustedSec.

Visit: https://www.trustedsec.com

It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

Select from the menu:

1) Social-Engineering Attacks
2) Penetration Testing (Fast-Track)
3) Third Party Modules
4) Update the Social-Engineer Toolkit
5) Update SET configuration
6) Help, Credits, and About

99) Exit the Social-Engineer Toolkit

set> 
```

Step 10: Now our SEToolkit has been downloaded on our system, it's time to use it. Now, we have to select the option from the following options. Option 2 is the one we've chosen.

```
File Actions Edit View Help
~/Desktop/SEToolkit/SEToolkit
.M""bgd 7MM""YMM MMP""MM""YMM
,MI "Y MM "7 P' MM "7
The Social-Engineer Toolkit (SET) MM by David Kennedy (ReL1K)
"YMMNg. MMmmMM MM
not running aMM MM Y MM
Mb dM MM ,M MM
Exiting "P"Ybmd" .JMMmmmmMMmm .JMMMLL!

[---] The Social-Engineer Toolkit (SET) [---]
[---] Created by: David Kennedy (ReL1K) [---]
Thank you for Version: 8.0.3 (Social-Engineer Toolkit)
Codename: 'Maverick'
[---] Follow us on Twitter: @TrustedSec [---]
[---] Follow me on Twitter: @HackingDave [---]
[---] Homepage: https://www.trustedsec.com [---]
Welcome to the Social-Engineer Toolkit (SET).
The one stop shop for all of your SE needs.

The Social-Engineer Toolkit is a product of TrustedSec.

Visit: https://www.trustedsec.com

It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

Select from the menu:

1) Spear-Phishing Attack Vectors
2) Website Attack Vectors
3) Infectious Media Generator
4) Create a Payload and Listener
5) Mass Mailer Attack
6) Arduino-Based Attack Vector
7) Wireless Access Point Attack Vector
8) QRCode Generator Attack Vector
9) Powershell Attack Vectors
10) Third Party Modules

99) Return back to the main menu.

set>
```

Website Attack Vectors:

1. Option 2

Step 11: Now, we are ready to set up a phishing page, we'll go with option 3, which is a credential harvester attack method.

1. option 3

The **Multi-Attack** method will add a combination of attacks through the web attack menu. For example you can utilize the Java Applet, Metasploit Browser, Credential Harvester/Tabnabbing all at once to see which is successful.

The **HTA Attack** method will allow you to clone a site and perform powershell injection through HTA files which can be used for Windows-based powershell exploitation through the browser.

```
~/Desktop/SToolkit/ctoolkit
$ ./ctoolkit.py
CToolkit v1.0
1) Java Applet Attack Method
2) Metasploit Browser Exploit Method
3) Credential Harvester Attack Method
4) Tabnabbing Attack Method
5) Web Jacking Attack Method
6) Multi-Attack Web Method (Toolkit/Metasploit)
7) HTA Attack Method
99) Return to Main Menu
$ ./ctoolkit.py -i http://10.10.10.10:8080/ctoolkit.py
~/Desktop/SToolkit/ctoolkit
set:webattack>
```

Step 12: Since we are making a phishing page; we'll go with option 1, which is a web template.

1. option 1

```
set:webattack>3

The first method will allow SET to import a list of pre-defined web
applications that it can utilize within the attack.

The second method will completely clone a website of your choosing
and allow you to utilize the attack vectors within the completely
same web application you were attempting to clone.

The third method allows you to import your own website, note that you
should only have an index.html when using the import website
functionality.

    1) Web Templates
    2) Site Cloner
    3) Custom Import

99) Return to Webattack Menu

set:webattack>1
[-] Credential harvester will allow you to utilize the clone capabilities within SET
[-] to harvest credentials or parameters from a website as well as place them into a rep
ort

----- * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * -----

The way that this works is by cloning a site and looking for form fields to
rewrite. If the POST fields are not usual methods for posting forms this
could fail. If it does, you can always save the HTML, rewrite the forms to
be standard forms and use the "IMPORT" feature. Additionally, really
important:

If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesn't know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perspective, it will not work. This isn't a SET issue
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [10.0.2.15]:
```

Step 13: The social engineering tool will now create a phishing page on our localhost.


```

File Actions Edit View Help
3) Custom Import~/Desktop/SETtoolkit/settoolkit/
99) Return to Webattack Menu

set:webattack>1
[~] Credential harvester will allow you to utilize the clone capabilities within SET
[~] to harvest credentials or parameters from a website as well as place them into a report

--- * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * ---

The way that this works is by cloning a site and looking for form fields to
rewrite. If the POST fields are not usual methods for posting forms this
could fail. If it does, you can always save the HTML, rewrite the forms to
be standard forms and use the "IMPORT" feature. Additionally, really
important:

If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesn't know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perspective, it will not work. This isn't a SET issue
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [10.0.2.15]:

**** Important Information ****

For templates, when a POST is initiated to harvest
credentials, you will need a site for it to redirect.

You can configure this option under:

/etc/settoolkit/set.config

Edit this file, and change HARVESTER_REDIRECT and
HARVESTER_URL to the sites you want to redirect to
after it is posted. If you do not set these, then
it will not redirect properly. This only goes for
templates.

1. Java Required
2. Google
3. Twitter

set:webattack> Select a template:

```

Step 14: Choose option 2 in order to create a Google phishing page, and a phishing page will be generated on our localhost.


```

set:webattack> IP address for the POST back in Harvester/Tabnabbing [10.0.2.15]:
.../settoolkit/settoolkit/webattack/

**** Important Information ****
The Social-Engineer Toolkit (SET) is a tool used to create phishing pages.
For templates, when a POST is initiated to harvest
credentials, you will need a site for it to redirect.
You can configure this option under:
/etc/settoolkit/set.config
Edit this file, and change HARVESTER_REDIRECT and
HARVESTER_URL to the sites you want to redirect to
after it is posted. If you do not set these, then
it will not redirect properly. This only goes for
templates.
.../settoolkit/settoolkit/webattack/

1. Java Required
2. Google
3. Twitter

set:webattack> Select a template:2
[*] Cloning the website: http://www.google.com
[*] This could take a little bit...
The best way to use this attack is if username and password form fields are available. R
egardless, this captures all POSTs on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:
.../settoolkit/settoolkit/webattack/

```

Step 15: A phishing page for Google is being created using the social engineering toolkit. As we can see, SEToolkit generate a phishing page of Google on our localhost (i.e., on our IP address). The social engineering toolset works in this manner. The social engineering toolkit will design our phishing page. Once the victim types the id password in the fields the id password will be shown on our terminal where SET is running.

